STRICT SIGN REGULARITY OF THE RECIPROCAL JACKSON q-GAMMA KERNEL

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Abstract. Let $0 < q < 1$ and let Γ_q be the Jackson q-gamma function. We prove that for every $r \geq 1$ and all real numbers $0 < x_1 < \cdots < x_r$ and $0 < y_1 < \cdots < y_r$ the determinant $\det[1/\Gamma_q(x_i + y_j)]_{i,j=1}^{r}$ is nonzero and has the sign $(-1)^{r(r-1)/2}$. Thus the kernel $(x, y) \mapsto 1/\Gamma_q(x + y)$ is strictly sign regular of every finite order on $(0, \infty)^2$, as conjectured by Karp, Vishnyakova and Zhang. The proof expands $(q^{x+y}; q)_\infty$ by the q-binomial theorem as a power series in q^y whose coefficients are polynomials in $1 - q^x$. A total nonnegativity theorem of Mainar, Peña and Rubio for the change of basis between monomial and Newton bases, combined with the sign of generalized Vandermonde determinants, shows that every term in the Cauchy–Binet expansion of a truncation has the predicted sign. The term of lowest degrees does not depend on the truncation and gives an explicit lower bound that persists in the limit. As a consequence, the discrete kernel $(x, n) \mapsto 1/(q^x; q)_n$ is strictly sign regular of every order, which extends a result of Karp, Vishnyakova and Zhang for orders at most three.

1. Introduction

Let $X, Y \subseteq \mathbb{R}$. Following Karlin [2, Chapter 2, §1] (see also [3, Definition 2.1]), a kernel $K\colon X \times Y \to \mathbb{R}$ is *sign regular of order* r, written $K \in SR_r$, if there are signs $\varepsilon_1, \ldots, \varepsilon_r \in \{1, -1\}$ such that

$$\varepsilon_m \det\big[K(x_i, y_j)\big]_{i,j=1}^{m} \geq 0 \tag{1.1}$$

for all $m \leq r$ and all $x_1 < \cdots < x_m$ in X and $y_1 < \cdots < y_m$ in Y. If the inequality in (1.1) is always strict, K is *strictly sign regular of order* r, written $K \in SSR_r$; if this holds for every r, we write $K \in SSR_\infty$. When all ε_m equal 1 one speaks of (strict) total positivity. Sign regular kernels are the natural setting for variation diminishing properties [2, Chapter 5].

Karp, Vishnyakova and Zhang [3] used kernels in SR_3 to show that ratios of functional series and of integral transforms preserve unimodality, and listed many kernels to which their theorems apply. For $0 < q < 1$ let

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \qquad (a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j),$$

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and let

$$\Gamma_q(z) = (1-q)^{1-z}\,\frac{(q;q)_\infty}{(q^z;q)_\infty}, \qquad z>0, \tag{1.2}$$

be the Jackson q-gamma function [5, §5.18(ii)]. In [3, Section 3, Examples 7 and 8] it is conjectured that the Hankel kernel $\Gamma_q(x+y)$ is strictly totally positive of every order and that the reciprocal kernel $1/\Gamma_q(x+y)$ belongs to SSR_∞. For the purposes of their unimodality theorem, Karp, Vishnyakova and Zhang proved the discrete statements that $(q^x;q)_n$ is strictly totally positive of order three, and that $1/(q^x;q)_n$ belongs to SSR_3 with the sign pattern $(+,-,-)$, on $(0,\infty)\times\mathbb{N}_0$ [3, Lemmas A.1 and A.2]. Salazar [6, Theorem 4.1] proved the first conjecture by writing Γ_q as the Mellin transform of a positive atomic measure. As he points out [6, Remark 4.2], the reciprocal kernel is not a positive moment kernel, and the second conjecture lies outside that method.

Throughout we write

$$\varepsilon_r = (-1)^{r(r-1)/2},$$

so that $(\varepsilon_1,\varepsilon_2,\varepsilon_3,\varepsilon_4,\dots) = (+,-,-,+,\dots)$. Our main result settles the second conjecture.

Theorem 1.1. *Let $0<q<1$ and $r\ge 1$. If $0<x_1<\cdots<x_r$ and $0<y_1<\cdots<y_r$, then*

$$\varepsilon_r \det\Big[\frac{1}{\Gamma_q(x_i+y_j)}\Big]_{i,j=1}^{r} > 0.$$

Consequently $(x,y)\mapsto 1/\Gamma_q(x+y)$ belongs to SSR_∞ on $(0,\infty)^2$, with the signs ε_r.

The proof gives an explicit lower bound, which remains valid when $x_1=0$.

Theorem 1.2. *Let $0<q<1$ and $r\ge 1$. If $0\le x_1<\cdots<x_r$ and $0<y_1<\cdots<y_r$, then*

$$\begin{aligned}\varepsilon_r \det\Big[\frac{1}{\Gamma_q(x_i+y_j)}\Big]_{i,j=1}^{r} \ge{}& \frac{q^{\binom{r}{3}}\,(1-q)^{\sum_i x_i+\sum_j y_j-r}}{(q;q)_\infty^{r}\,\prod_{k=0}^{r-1}(q;q)_k} \\ &\times \prod_{j=1}^{r}(q^{y_j};q)_\infty \prod_{1\le i<j\le r}(q^{x_i}-q^{x_j})(q^{y_i}-q^{y_j}).\end{aligned} \tag{1.3}$$

The right-hand side is positive.

Since the kernel is symmetric in x and y and transposition does not change determinants, the roles of (x_i) and (y_j) in Theorem 1.2 may be interchanged. The case $x_1=0$ extends the strict sign regularity of the discrete kernel in [3, Lemma A.2] from order three to all orders.

Corollary 1.3. *Let $0<q<1$ and $r\ge 1$. If $0<\mu_1<\cdots<\mu_r$ are real and $0\le n_1<\cdots<n_r$ are integers, then*

$$\varepsilon_r \det\Big[\frac{1}{(q^{\mu_j};q)_{n_i}}\Big]_{i,j=1}^{r} > 0.$$

Thus $(\mu,n)\mapsto 1/(q^{\mu};q)_n$ belongs to SSR_∞ on $(0,\infty)\times\mathbb{N}_0$, with the signs ε_r. For $r\le 3$ this is [3, Lemma A.2].

Method. Up to positive row and column factors, $1/\Gamma_q(x+y)$ equals $(q^{x+y};q)_\infty$. With $u=q^x$ and $v=q^y$, the q-binomial theorem gives

$$\frac{(uv;q)_\infty}{(v;q)_\infty}=\sum_{k=0}^{\infty}\frac{(u;q)_k}{(q;q)_k}\,v^k.$$

As a polynomial in $s=1-u$, the coefficient $(u;q)_k$ is a positive multiple of $\prod_{j<k}(s+q^{-j}-1)$, which is, up to the sign $(-1)^k$, the Newton polynomial with the nonnegative nodes $q^{-j}-1$ evaluated at $-s$. Mainar, Peña and Rubio [4] obtained bidiagonal factorizations of the change of basis matrices between monomial and Newton bases and characterized their total positivity in terms of the signs of the nodes. Lemma 3.1 translates their Corollary 5 into the present notation: the coefficient matrix of these polynomials is totally nonnegative. Combined with the sign of generalized Vandermonde determinants, the Cauchy–Binet formula shows that every term in the expansion of a truncated determinant becomes nonnegative after multiplication by ε_r. The term indexed by the degrees $0,1,\dots,r-1$ is the same for every truncation and is an explicit product of positive factors; it gives the bound (1.3). The truncated matrices converge entrywise, so their determinants converge by continuity, and the bound persists in the limit.

Dependencies. Besides the Cauchy–Binet formula, the proof uses two known results that we do not reprove: the q-binomial theorem, and [4, Corollary 5] for increasing nonnegative nodes. Theorem 1.1 is obtained by applying these ingredients to the reciprocal q-gamma kernel. To our knowledge, Theorem 1.1 has not appeared in the literature.

Organization. Section 2 recalls the q-binomial theorem and a sign lemma for generalized Vandermonde determinants. Section 3 treats the coefficient polynomials. Section 4 proves Theorems 1.1 and 1.2 and Corollary 1.3. Section 5 records a consequence for the classical gamma function.

2. Preliminaries

Throughout, $0<q<1$ is fixed. If $a<1$, every factor of $(a;q)_\infty$ is positive, and the product converges to a positive number because $\sum_j |a|q^j<\infty$. In particular $(q;q)_\infty>0$ and $(q;q)_k\ge(q;q)_\infty$ for all $k\ge 0$.

Lemma 2.1 (q-binomial theorem)**.** *For every real a and every real z with $|z|<1$,*

$$\sum_{k=0}^{\infty}\frac{(a;q)_k}{(q;q)_k}\,z^k=\frac{(az;q)_\infty}{(z;q)_\infty},$$

and the series converges absolutely.

Proof. This is [5, (17.5.2)], with the series definition [5, (17.4.1)]; see also [1, Section 1.3]. Absolute convergence follows from $|(a;q)_k|\le(-|a|;q)_\infty$ and $(q;q)_k\ge(q;q)_\infty>0$. □

For matrices we use the following notation. If F has r rows and K is a set of r column indices, $F[\,\cdot\,,K]$ is the submatrix with these columns; similarly $G[K,\cdot\,]$ consists of the rows of G indexed by K, and $F[I,K]$ has rows I and columns K,

all in increasing order. The Cauchy–Binet formula states that if F is $r \times N$ and G is $N \times r$, then

$$\det(FG) = \sum_{K} \det F[\,\cdot\,, K]\, \det G[K, \cdot\,], \tag{2.1}$$

the sum running over all r-element subsets K of the index set of size N.

The next lemma is classical [2, Chapter 2]; we include the short argument.

Lemma 2.2. *Let $0 < t_1 < \cdots < t_r$ and let $\alpha_1 < \cdots < \alpha_r$ be real. Then $\det[t_i^{\alpha_j}]_{i,j=1}^{r} > 0$. If instead $0 = t_1 < t_2 < \cdots < t_r$ and $0 \le \alpha_1 < \cdots < \alpha_r$, with $0^0 = 1$, then $\det[t_i^{\alpha_j}]_{i,j=1}^{r} \ge 0$.*

Proof. We first show that a function $f(t) = \sum_{j=1}^{m} a_j t^{\beta_j}$ with distinct real exponents and nonzero coefficients has at most $m-1$ zeros in $(0,\infty)$. For $m = 1$ this is clear. For $m \ge 2$ we may assume $\beta_1 < \cdots < \beta_m$; then the derivative of $t^{-\beta_1} f(t)$ is $\sum_{j=2}^{m} a_j(\beta_j - \beta_1) t^{\beta_j - \beta_1 - 1}$, which by induction has at most $m-2$ positive zeros, and Rolle's theorem gives at most $m-1$ positive zeros of f.

If $\det[t_i^{\alpha_j}] = 0$ for some $0 < t_1 < \cdots < t_r$, a nonzero vector $(a_1, \dots, a_r)$ satisfies $\sum_j a_j t_i^{\alpha_j} = 0$ for all i; discarding the vanishing a_j contradicts the bound just proved. Hence the determinant does not vanish on the convex set $\{0 < t_1 < \cdots < t_r\}$, and by continuity its sign is constant there. Take $t_i = \lambda^i$ with $\lambda > 1$. Then

$$\det[t_i^{\alpha_j}] = \sum_{\sigma} \operatorname{sgn}(\sigma)\, \lambda^{\sum_i i\,\alpha_{\sigma(i)}},$$

and since both i and α_i are strictly increasing, the exponent $\sum_i i\,\alpha_{\sigma(i)}$ is uniquely maximized by the identity permutation. Hence the determinant is positive for large λ, and therefore everywhere.

If $t_1 = 0$, the first row is $(1, 0, \dots, 0)$ when $\alpha_1 = 0$ and vanishes when $\alpha_1 > 0$. Expanding along it, the determinant is either 0 or the determinant for the nodes $t_2, \dots, t_r$ and exponents $\alpha_2, \dots, \alpha_r$, which is positive by the first part. □

3. Newton coefficient polynomials

Put $c_j = q^{-j} - 1$ for $j \ge 0$, so that $0 = c_0 < c_1 < c_2 < \cdots$, and define

$$a_k(s) = \prod_{j=0}^{k-1} (s + c_j) = \sum_{d=0}^{k} A_{k,d}\, s^d, \qquad p_k(s) = q^{\binom{k}{2}} a_k(s) \qquad (k \ge 0).$$

Since $1 - (1-s)q^j = q^j(s + c_j)$ and $\sum_{j<k} j = \binom{k}{2}$, we have

$$p_k(1-u) = (u;q)_k \qquad \text{for all real } u. \tag{3.1}$$

A matrix is *totally nonnegative* if all its minors are nonnegative; in [4] such matrices are called totally positive.

Lemma 3.1 (Mainar–Peña–Rubio)**.** *Let $M \ge 0$, and let $A^{(M)}$ be the matrix with entries $A_{k,d}$, where $0 \le k, d \le M$. Then $A^{(M)}$ is lower triangular with unit diagonal, and it is totally nonnegative.*

Proof. The triangular structure and the unit diagonal are clear, since a_k is monic of degree k. For the total nonnegativity we translate [4, Corollary 5]; we index rows and columns from 0 instead of 1. Let $t_0 \le t_1 \le \cdots \le t_{M-1}$ be nodes, let $w_k(t) = \prod_{j<k}(t - t_j)$ be the Newton basis and $m_d(t) = t^d$ the monomial basis of

the polynomials of degree at most M, and let U be the upper triangular matrix with

$$(m_0, \dots, m_M) = (w_0, \dots, w_M)\, U,$$

see [4, (19)–(20)]. With $J = \operatorname{diag}((-1)^k)_{0 \le k \le M}$, [4, Corollary 5] states that $JU^{-1}J$ is totally nonnegative when all nodes are nonnegative.

Take $t_j = c_j$, which are nonnegative and increasing. Since $(w_0, \dots, w_M) = (m_0, \dots, m_M)U^{-1}$, we have $w_k(t) = \sum_d (U^{-1})_{d,k} t^d$, and therefore

$$a_k(s) = (-1)^k w_k(-s) = \sum_d (-1)^{k+d} (U^{-1})_{d,k}\, s^d.$$

Thus $A_{k,d} = (JU^{-1}J)_{d,k}$, that is, $A^{(M)} = (JU^{-1}J)^T$, and the transpose of a totally nonnegative matrix is totally nonnegative. □

Remark 3.2. The total nonnegativity in Lemma 3.1 can also be read off from [4, Theorem 3]. By [4, (24)–(25)], $U^{-1} = G_1' \cdots G_M'$, where each G_i' is upper bidiagonal with unit diagonal and with off-diagonal entries of the form $-t_l$. Conjugation by J changes the sign of the off-diagonal entries of a bidiagonal matrix, so $JU^{-1}J = (JG_1'J) \cdots (JG_M'J)$ is a product of upper bidiagonal matrices whose off-diagonal entries are nodes $c_l \ge 0$. Since transposition reverses the order of the factors,

$$A^{(M)} = (JU^{-1}J)^T = (JG_M'J)^T \cdots (JG_1'J)^T$$

is a product of lower bidiagonal matrices with nonnegative entries.

Proposition 3.3. *Let $0 \le s_1 < \cdots < s_r$ be real.*

(a) *For all integers $0 \le k_1 < \cdots < k_r$,*

$$\det\big[p_{k_j}(s_i)\big]_{i,j=1}^{r} \ge 0.$$

(b) *For the consecutive degrees $0, 1, \dots, r-1$,*

$$\det\big[p_{j-1}(s_i)\big]_{i,j=1}^{r} = q^{\binom{r}{3}} \prod_{1 \le i < j \le r} (s_j - s_i) > 0.$$

Proof. Choose $M \ge k_r$, let $V = (s_i^d)$ be the $r \times (M+1)$ matrix with $1 \le i \le r$ and $0 \le d \le M$, and let $C = (q^{\binom{k}{2}} A_{k,d})_{0 \le k,d \le M}$. Then $p_k(s) = \sum_d C_{k,d} s^d$, so with $K = \{k_1, \dots, k_r\}$ we have $[p_{k_j}(s_i)] = V\,(C[K, \cdot])^T$, and (2.1) gives

$$\det\big[p_{k_j}(s_i)\big] = \sum_D \det V[\,\cdot\,, D]\, \det C[K, D],$$

where D runs over the r-element subsets of $\{0, \dots, M\}$. Every $\det V[\,\cdot\,, D]$ is nonnegative by Lemma 2.2, and $\det C[K, D] = q^{\sum_{k \in K} \binom{k}{2}} \det A^{(M)}[K, D] \ge 0$ by Lemma 3.1. This proves (a).

For (b), the polynomial p_k has degree k and leading coefficient $q^{\binom{k}{2}}$. Subtracting from each column suitable multiples of the preceding columns reduces the matrix to $[q^{\binom{j-1}{2}} s_i^{j-1}]$, whose determinant is $q^{\sum_{k=0}^{r-1} \binom{k}{2}}$ times a Vandermonde determinant. Since $\sum_{k=0}^{r-1} \binom{k}{2} = \binom{r}{3}$, the formula follows. □

If $s_1 > 0$, the term $D = K$ in the proof of (a) is positive, because $C[K, K]$ is lower triangular with positive diagonal; hence the inequality in (a) is then strict. We shall not need this.

4. Proof of the main results

Proposition 4.1. *Let $r \geq 1$, $1 \geq u_1 > \cdots > u_r > 0$ and $1 > v_1 > \cdots > v_r > 0$. Then*

$$\varepsilon_r \det\big[(u_i v_j;q)_\infty\big]_{i,j=1}^r \geq \frac{q^{\binom{r}{3}} \prod_{j=1}^r (v_j;q)_\infty}{\prod_{k=0}^{r-1}(q;q)_k} \prod_{1\leq i<j\leq r} (u_i-u_j)(v_i-v_j) > 0.$$

Proof. Put $s_i = 1-u_i$, so that $0 \leq s_1 < \cdots < s_r < 1$ and $s_j - s_i = u_i - u_j$. For $M \geq r-1$ let H_M be the $r \times r$ matrix with entries

$$H_M(i,j) = \sum_{k=0}^{M} \frac{p_k(s_i)}{(q;q)_k}\, v_j^k.$$

Then $H_M = P\,\Delta\, W$, where $P = (p_k(s_i))$ is $r \times (M+1)$, $\Delta = \operatorname{diag}(1/(q;q)_k)_{0\leq k\leq M}$ and $W = (v_j^k)$ is $(M+1)\times r$. By (2.1),

$$\det H_M = \sum_K \det P[\,\cdot\,, K] \Big(\prod_{k\in K} \frac{1}{(q;q)_k}\Big) \det W[K,\cdot\,], \tag{4.1}$$

the sum running over the r-element subsets $K = \{k_1 < \cdots < k_r\}$ of $\{0,\dots,M\}$.

We determine the signs of the factors. By Proposition 3.3(a), $\det P[\,\cdot\,, K] \geq 0$. Next, $\det W[K,\cdot\,] = \det[v_j^{k_l}]_{l,j=1}^r$ equals the determinant of its transpose $[v_j^{k_l}]_{j,l=1}^r$, whose rows correspond to the decreasing nodes $v_1 > \cdots > v_r$. Reversing the order of the rows, a permutation of sign ε_r, produces a matrix with increasing positive nodes and increasing exponents, whose determinant is positive by Lemma 2.2. Hence $\varepsilon_r \det W[K,\cdot\,] > 0$, and every term of $\varepsilon_r \det H_M$ in (4.1) is nonnegative.

Keep only the term $K_0 = \{0,1,\dots,r-1\}$, which occurs for every $M \geq r-1$. By Proposition 3.3(b) and the Vandermonde formula $\det[v_j^{l-1}]_{l,j=1}^r = \prod_{i<j}(v_j - v_i)$,

$$\varepsilon_r \det H_M \geq \frac{q^{\binom{r}{3}}}{\prod_{k=0}^{r-1}(q;q)_k} \prod_{1\leq i<j\leq r} (u_i-u_j)(v_i-v_j) \qquad (M \geq r-1). \tag{4.2}$$

The right-hand side is positive and does not depend on M.

By (3.1) and Lemma 2.1 with $a = u_i$ and $z = v_j$,

$$\lim_{M\to\infty} H_M(i,j) = \sum_{k=0}^{\infty} \frac{(u_i;q)_k}{(q;q)_k}\, v_j^k = \frac{(u_i v_j;q)_\infty}{(v_j;q)_\infty}$$

for all i,j. The determinant of an $r\times r$ matrix is a polynomial in its entries, so $\det H_M$ converges to $\det[(u_i v_j;q)_\infty/(v_j;q)_\infty]$, and (4.2) passes to the limit. Multiplying column j by $(v_j;q)_\infty > 0$ for each j gives the assertion. □

Proof of Theorem 1.2. Put $u_i = q^{x_i}$ and $v_j = q^{y_j}$. Since $0 \leq x_1 < \cdots < x_r$ and $0 < y_1 < \cdots < y_r$, we have $1 \geq u_1 > \cdots > u_r > 0$ and $1 > v_1 > \cdots > v_r > 0$. By (1.2),

$$\frac{1}{\Gamma_q(x_i+y_j)} = \frac{(1-q)^{x_i}\,(1-q)^{y_j}}{(1-q)\,(q;q)_\infty}\,(u_i v_j;q)_\infty.$$

Taking out the positive row factors $(1-q)^{x_i}$, the positive column factors $(1-q)^{y_j}$ and the positive scalar $((1-q)(q;q)_\infty)^{-1}$ common to all entries, we obtain

$$\det\Big[\frac{1}{\Gamma_q(x_i+y_j)}\Big] = \frac{(1-q)^{\sum_i x_i + \sum_j y_j}}{(1-q)^r (q;q)_\infty^r} \det\big[(u_i v_j;q)_\infty\big],$$

and Proposition 4.1 gives (1.3). Every factor on the right-hand side of (1.3) is positive: $(q^{y_j};q)_\infty > 0$ because $q^{y_j} < 1$, and $q^{x_i} > q^{x_j}$, $q^{y_i} > q^{y_j}$ for $i < j$. For $r = 1$ all empty products equal 1. □

Proof of Theorem 1.1. This is the special case $x_1 > 0$ of Theorem 1.2. Since the bound holds for every $r \geq 1$, the kernel belongs to SSR_∞ with the signs ε_r. □

Proof of Corollary 1.3. By (1.2), for $\mu > 0$ and $n \in \mathbb{N}_0$,

$$\frac{\Gamma_q(\mu+n)}{\Gamma_q(\mu)} = (1-q)^{-n}\,\frac{(q^\mu;q)_\infty}{(q^{\mu+n};q)_\infty} = (1-q)^{-n}(q^\mu;q)_n,$$

so that

$$\frac{1}{(q^\mu;q)_n} = (1-q)^{-n}\,\Gamma_q(\mu)\,\frac{1}{\Gamma_q(n+\mu)}.$$

The factors $(1-q)^{-n}$ and $\Gamma_q(\mu)$ are positive row and column factors. Apply Theorem 1.2 with $x_i = n_i$ and $y_j = \mu_j$; the case $n_1 = 0$ is allowed there. □

5. A remark on the classical gamma function

Remark 5.1. In [3, Section 3, Example 6] it is conjectured, on the basis of numerical evidence, that $1/\Gamma(x+y)$ belongs to SSR_∞ on $(0,\infty)^2$. Since $\Gamma_q(z) \to \Gamma(z)$ as $q \to 1^-$ for every $z > 0$ [1, Section 1.10], the matrices in Theorem 1.1 converge entrywise, and hence their determinants converge. We obtain

$$\varepsilon_r \det\Big[\frac{1}{\Gamma(x_i+y_j)}\Big]_{i,j=1}^{r} \geq 0$$

for all $r \geq 1$, $0 < x_1 < \cdots < x_r$ and $0 < y_1 < \cdots < y_r$. Thus $1/\Gamma(x+y)$ is sign regular of every order on $(0,\infty)^2$, with the signs ε_r. This limiting argument establishes sign regularity only; strict sign regularity of $1/\Gamma(x+y)$ would require a separate argument.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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